

Khadeeja Tapkirwala

Houston, Texas

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Education

University of Houston

January 2024 – December 2027

Bachelor of Science in Computer Science, Data Science Concentration

Houston, Texas

- Microsoft Technical Resilience Certification, Spring 2025
- Relevant Coursework: OOP, Linear Algebra, Data Structures and Algorithms, Statistics for the Sciences, Principles of Data Management, Introduction to Machine Learning

Technical Skills

Languages: Python, R, C++, JavaScript, TypeScript, HTML, CSS, SQL

Libraries/Frameworks: NumPy, Pandas, OpenCV, PyTorch, MediaPipe, React

Tools: Linux, VS Code, Git, GitHub, Replit, Jupyter Notebook, Figma, Power BI, Tableau, MS Office

Work Experience

Structures & Artificial Intelligence Lab (SAIL), UH

January 2026 – Present

Research Assistant

Houston, Texas

- Developing state-of-the-art computer vision-based artificial intelligence solutions as part of an AI research team.
- Following current best practices in deep learning workflows, from data generation and preprocessing to model training and validation.

Self Employed

August 2023 – Present

Arabic Instructor

Houston, Texas

- Managed a portfolio of eight long term clients, delivering personalized instruction and maintaining 100% retention through structured learning plans.

Modest Essentials, LLC

June 2024 – December 2025

Business Operations Assistant

Stafford, Texas

- Supported operations and stakeholder communication for a small business, coordinating logistics across 150+ product designs.

Leadership Experience

Cougarettes (ACM-W Chapter at UH)

January 2026 – Present

Vice President

Houston, Texas

- Collaborated with 10+ student organizations and company sponsors to organize mentorship programs and hackathons.

Cougarettes (ACM-W Chapter at UH)

September 2025 – December 2025

Coding Event Chair, Microsoft Tech Resilience Facilitator

Houston, Texas

- Led and coordinated 15+ coding workshops on HTML/CSS, JavaScript, C++, and Python to strengthen peers technical skills.
- Planned and conducted three Microsoft Technical Resilience sessions to build peers soft skills.

Project Experience

Depth Refinement Neural Network | SAIL, UH | Python, PyTorch, OpenCV

January 2026 - Present

- Developing an AI model that enhances depth data from sensors using deep learning, implementing and training the model in PyTorch to contribute to research efforts.

Mindful Market | CodeRed Hackathon | React, TypeScript, Gemini API, Tailwind

October 2025

- Collaboratively built a sustainability web application in 24 hours, integrating Google Gemini via structured prompt engineering to produce automated eco scores and carbon impact evaluations.

Driver Drowsiness Detection System | Python, OpenCV, MediaPipe, NumPy

June 2025

- Developed an AI-driven monitoring system leveraging data-driven insights for real-time decision-making.

Students for Refugees Website | 1st Place, UH CodeJam | JavaScript, HTML/CSS

March 2025

- Collaborated in a four person team to design, prototype, and deploy a client specified website within four hours using Figma and Replit, winning first place among three competing teams.